Aarav Jain

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EDUCATION

Purdue University | Purdue Engineering — **West Lafayette, IN**
Candidate for Bachelor of Science in Computer Engineering — Expected *May 2028*

SKILLS

Languages: Python, HTML/CSS/JavaScript, C/C++, Java, SQL
Technologies: React, TypeScript, React Native, Docker, AWS, Git, NextJS, Langchain, NumPy, FastAPI, IsaacSim, Zephyr

WORK & LEADERSHIP EXPERIENCE

Algoverse
Machine Learning Researcher — Aug. 2024 – May 2025

- Developed and deployed a novel evaluation framework using Python and statistical analysis to quantify sycophantic behavior in multi-turn LLM conversations, identifying a **47% accuracy decline** over extended interactions
- Engineered an end-to-end testing pipeline leveraging OpenAI, Llama, and Claude APIs; integrated TruthfulQA and MMLU datasets from Huggingface to test **5k+ conversation samples**, measuring accuracy degradation and sycophantic tendencies
- Co-authored a research paper documenting the "Truth Decay" methodology, experimental pipeline, findings, and best-practice recommendations; **Published at the NAACL 2025 Student Research Workshop (SRW)**

Embedded Systems
Member — Sep. 2025 – Present

- Developing a Hardware-in-the-Loop (HiL) with multi-threaded architecture managing flight control, landing, autonomous hover, and controlled flight threads that process simulated sensor data to validate drone firmware in a safe environment.
- Engineering custom device drivers in Zephyr RTOS for IMU and 1D LiDAR sensors using I2C communication protocols, implementing low-level hardware interfaces to enable real-time sensor data simulation and firmware validation.
- Building a multi-processor communication framework with an orchestrator compiling firmware and generating sensor instructions, a test node performing sensor injection, and a drone processing simulated inputs to validate flight outputs.

Purdue IDEAS Research Lab
Undergraduate Researcher — Oct. 2025 – Present

- Developing an autonomous UAV-UGV natural language collaboration system using LLMs, VLMS, and computer vision to enable navigation and goal-oriented behavior in unstructured environments, leveraging terrain for strategic movement
- Implementing intelligent control algorithms using reinforcement learning to improve UAV-UGV communication for cooperative path planning to determine the optimal UGV navigation under hostile or uncertain terrain condition

PROJECTS

Marvel Oracle | Python, Typescript, NextJS, Pinecone, Langchain, FastAPI — github.com/Aarav-J/marvel_rag

- Implemented a chatbot with accurate information, context-aware responses, persistent chat history, and multi-chat sessions
- Embedded **9k Marvel Wiki articles into a Pinecone vector** index via a BeautifulSoup4 pipeline using automated parsing
- Architected a Retrieval Augmented Generation (RAG) platform using Langchain and Pinecone with dense search, metadata filtering, and top-k tuning for real-time chat with **30% increased answer relevance** compared to regular LLMs
- Developed and deployed a NextJS application with a FastAPI backend to deliver an interactive Marvel chatbot experience

Bridge | Next.js, ReactJS, TypeScript, Socket.IO, Express, WebRTC, Supabase — github.com/Aarav-J/bridge

- Designed and implemented a structured turn-based video debate system, matching users across opposite ends of the political spectrum using a quiz-based profiling algorithm and real-time topic selection, to enforce civility in debate.
- Built the backend infrastructure using Socket.IO/WebRTC for real-time communication and cross-spectrum matchmaking.
- Developed the front-end architecture using Next.JS, React, and Typescript to create a platform for onboarding and debates

RouteNote | Python, Yolo, NextJS, Typescript, OpenCV, FastAPI — github.com/Aarav-J/routenote

- Trained a **YOLOV11 model** in Python to detect climbing holds on wall images, outputting precise bounding boxes.
- Annotated 100+ climbing wall images for hold detection, creating a labeled dataset that improved route classifier accuracy
- Built a YOLO-based color classifier that filters holds by HSV color and confidence scores enabling precise route detection
- Developed a Next.JS + Firebase frontend integrated with the route detection API for user notetaking on routes and holds